Module 09: Active Directory Penetration Testing

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# Exercise 03: Performing RPC Client Enumeration to Collect Information about Active Directory Envi
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Scenario

Remote Procedure Call (RPC) interfaces exposed on Windows Domain Controllers (`MSRPC` on TCP port `135` / SMB named pipes `445`) allow authenticated users to perform low-level administrative and directory queries using the `rpcclient` utility. Through RPC calls, penetration testers can enumerate domain user account Relative Identifiers (RIDs), privilege tokens, group memberships, domain password policies, and network shares without relying on complex GUI tools or noisy scanning suites.

In this exercise, we utilize `rpcclient` from **Parrot Security** to interact with the target Windows Server 2019 Domain Controller (**192.168.177.19**). We perform user enumeration (`enumdomusers`), inspect token privileges (`enumprivs`), query user account properties via RID (`queryuser 0xc3a` , `queryuser 0x1f4`), discover domain groups (`enumdomgroups`), enumerate SMB shares (`netshareenumall`), extract domain password policy (`getusrdompwinfo 0x1f4`), and retrieve the primary Domain Controller name (`getdcname LPT`).

Objective

The objective of this lab is to help students learn how to:
- Establish an authenticated RPC connection to target Domain Controller **192.168.177.19** using **rpcclient -U Hand 192.168.177.19**
- Enumerate all domain accounts (**enumdomusers**) and extract user RIDs (e.g., `jdoe` RID: **[0xe3c]**)
- Inspect user privileges and delegation rights (**enumprivs**)
- Query detailed account metadata for specific RIDs using **queryuser 0xc3a** (user `Hand`) and **queryuser 0x1f4** (`Administrator`)
- Enumerate domain structures (**enumdomains**) and domain group memberships (**enumdomgroups**)
- Connect as Domain Administrator using **rpcclient -U Administrator 192.168.177.19**
- Enumerate all active SMB network shares using **netshareenumall**
- Extract domain password complexity policy via **getusrdompwinfo 0x1f4**
- Discover the primary Domain Controller netbios/DNS name using **getdcname LPT**

Task 1: Execute RPC Client Enumeration to Extract Domain Users, RIDs, Shares, and Password Policies

In this task, we connect to target Domain Controller **192.168.177.19** via `rpcclient` using domain user **Hand** (**Password3**) and **Administrator** (**Pa$$w0rd**), enumerating RIDs, privileges, shares, domain password policies, and DC details.

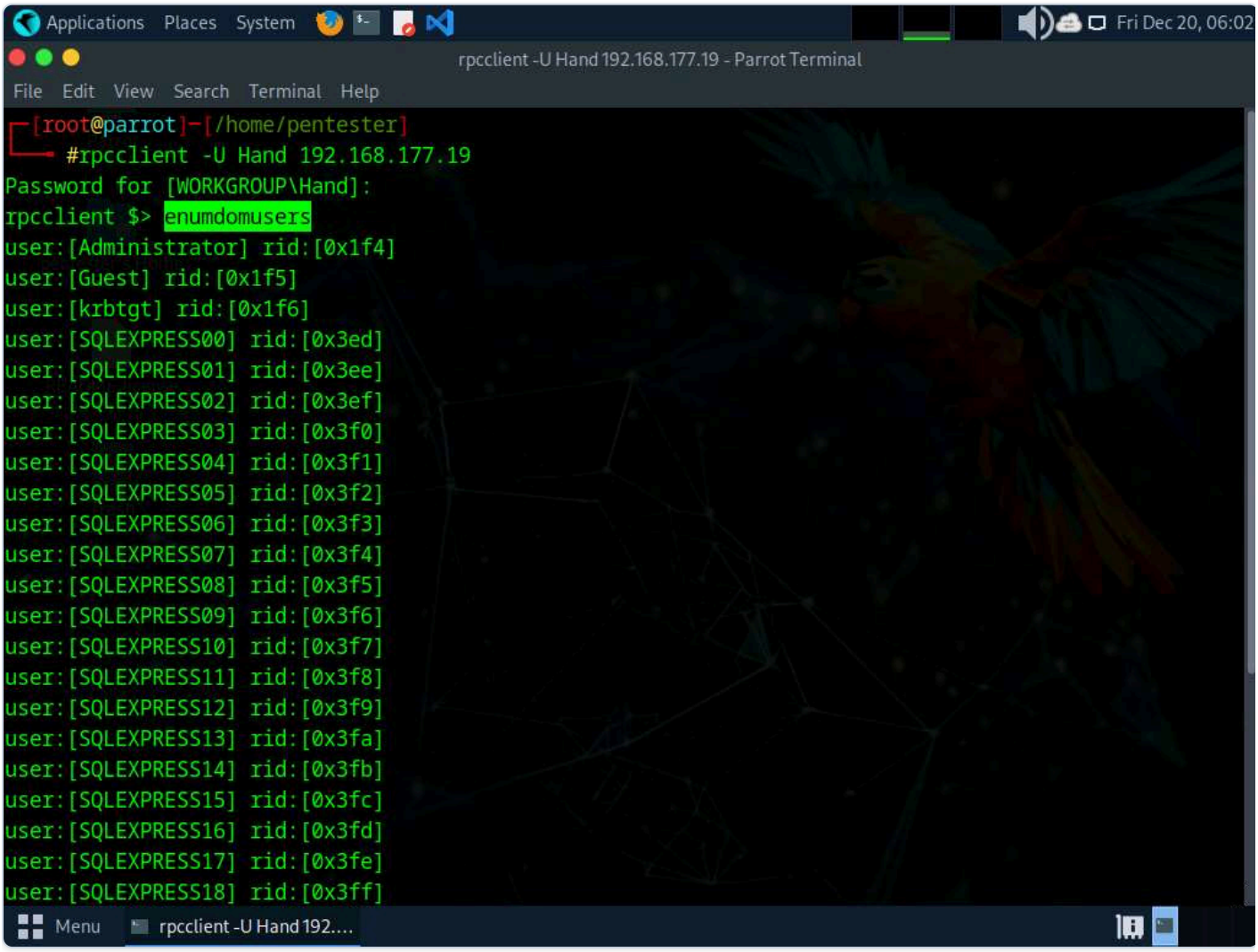
Lab Tasks

1. Switch to **Parrot Security** machine and log in (**pentester** / **toor**). Open terminal.
2. Establish an RPC client session with target machine **192.168.177.19** using domain credentials (**Hand** / **Password3**): **rpcclient -U Hand 192.168.177.19**

3. Enumerate all domain user accounts and their associated Relative Identifiers (RIDs):

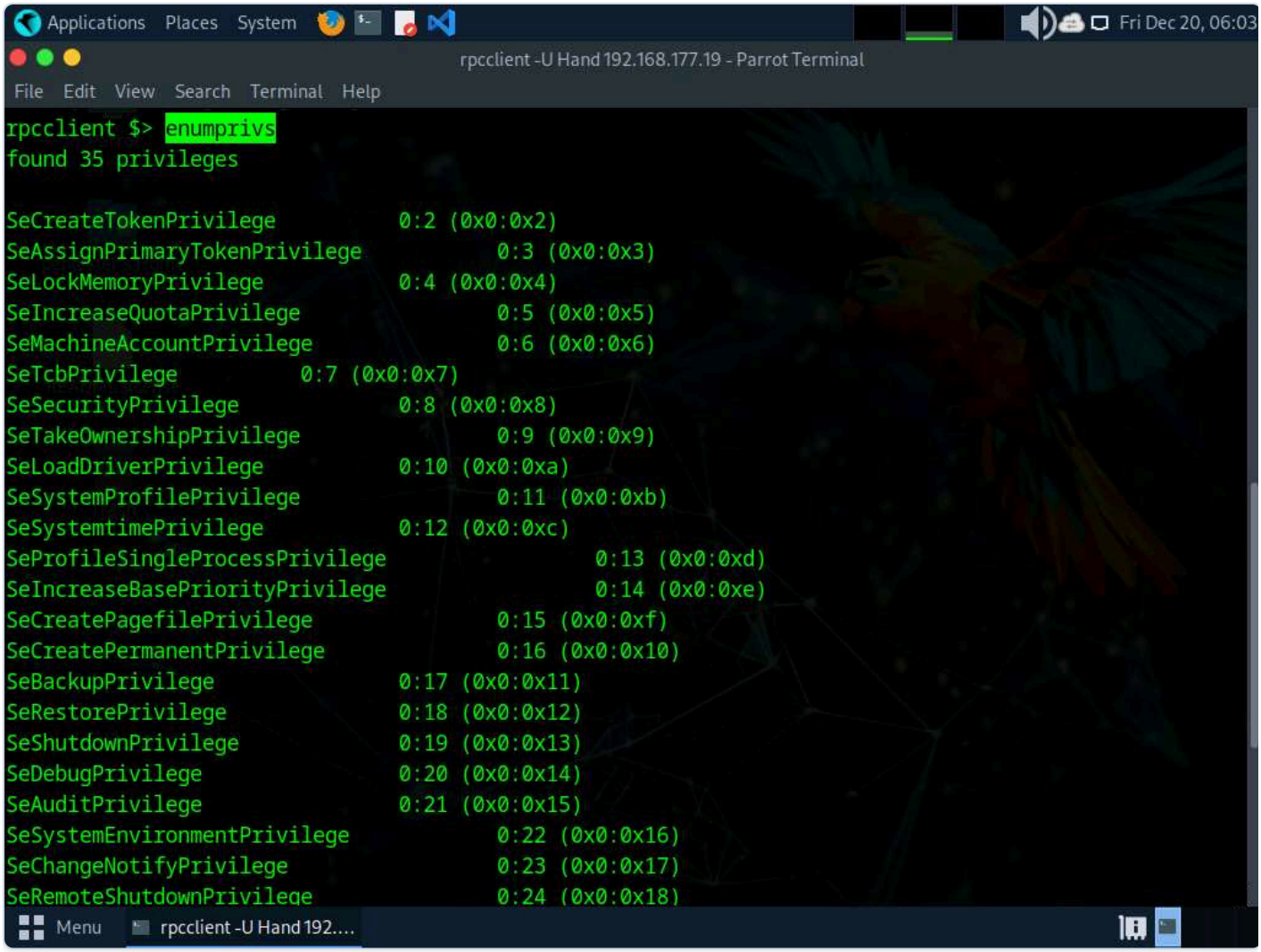
   enumdomusers

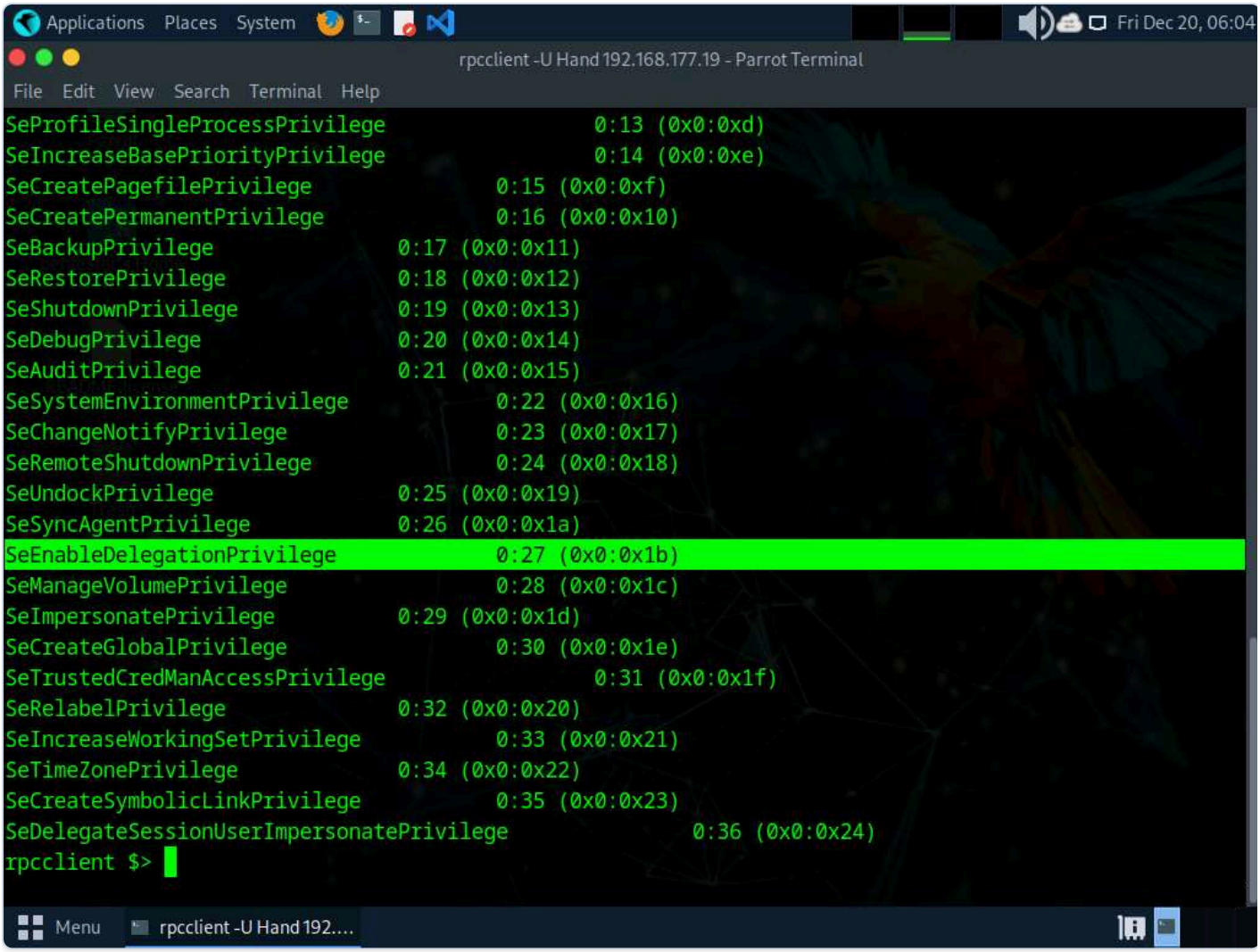
   Observe domain user `jdoe` *with RID:* ***[0xe3c]***.



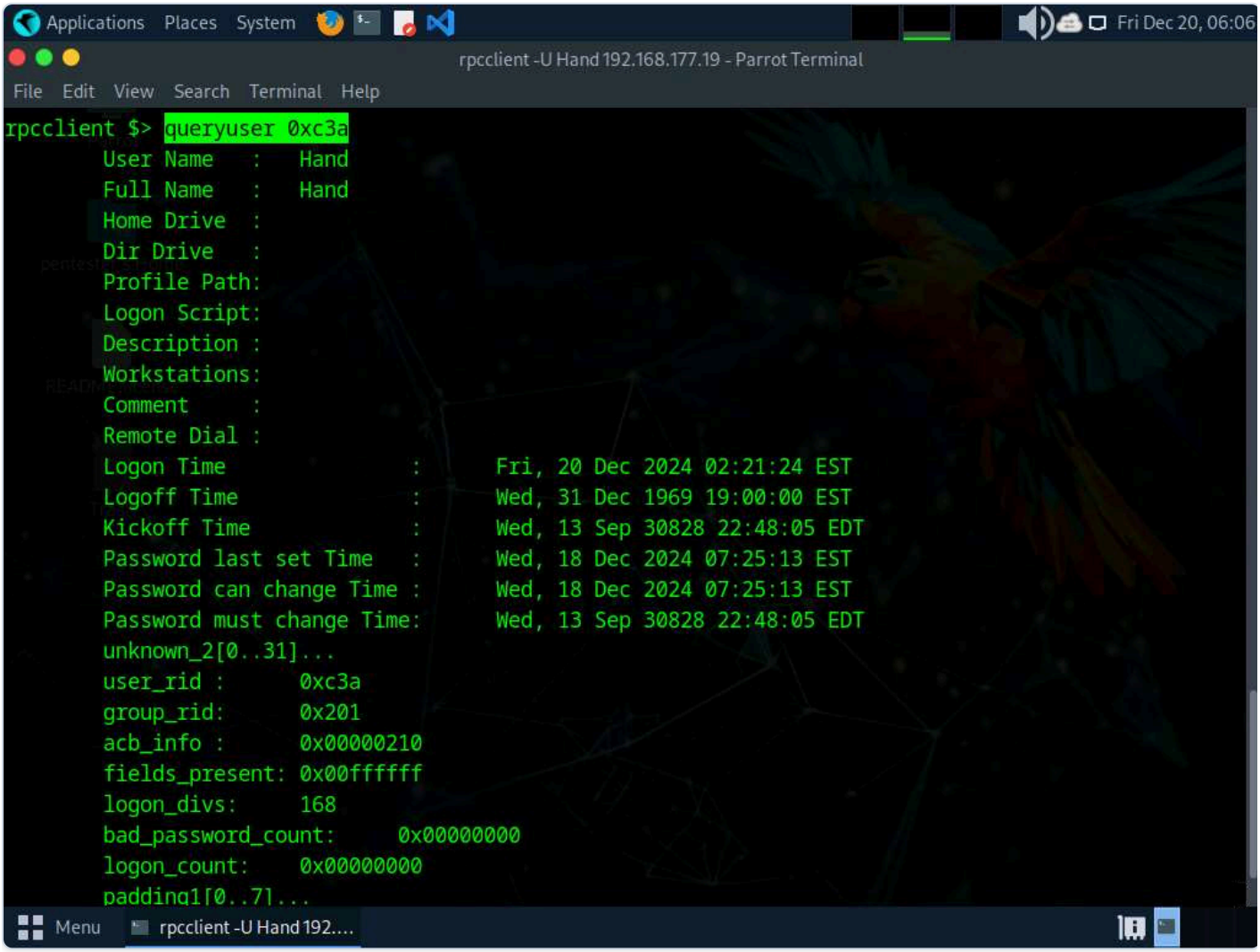
1. Enumerate current user rights and delegation privileges:

   enumprivs

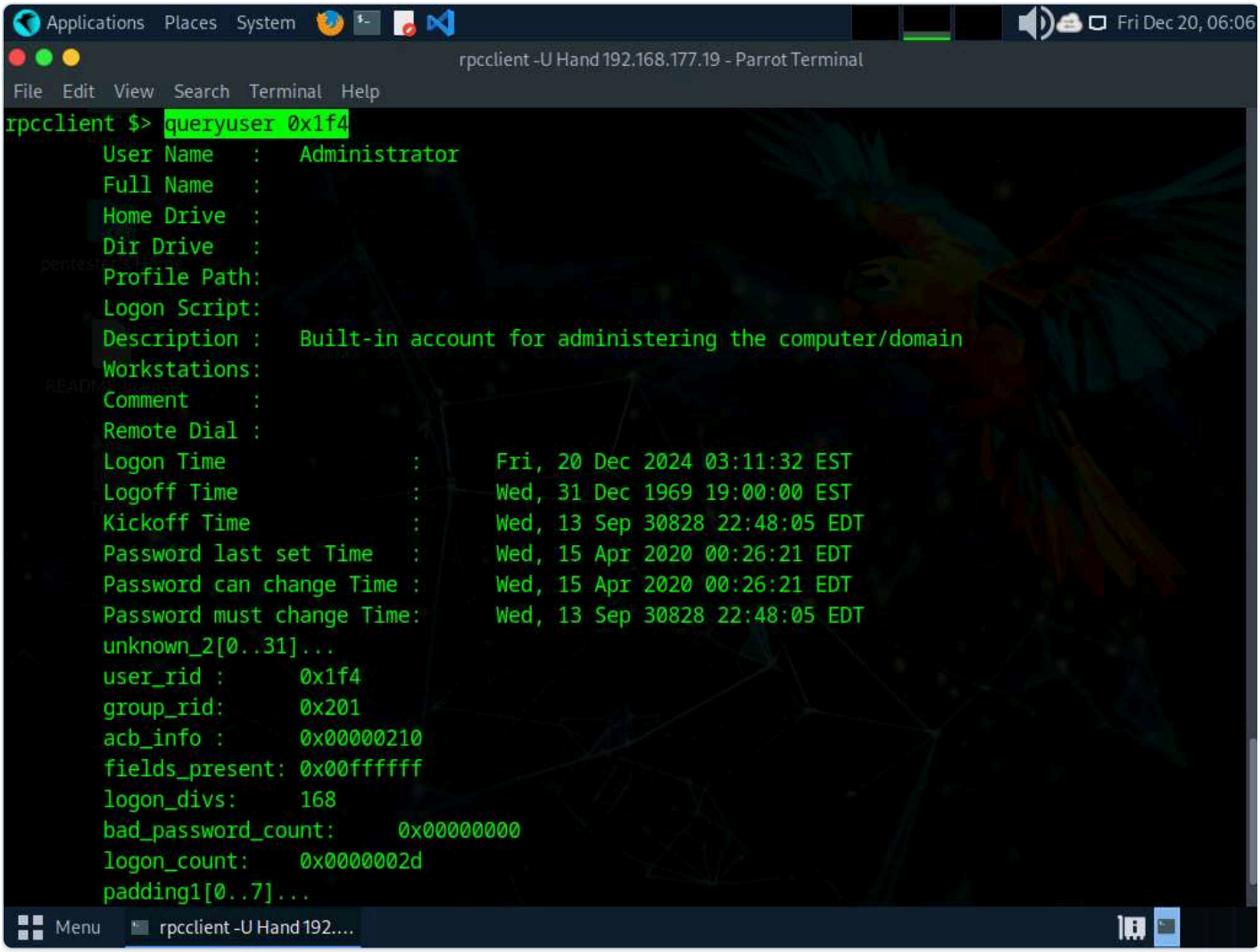

Applications Places System
Fri Dec 20, 06:03
rpcclient -U Hand 192.168.177.19 - Parrot Terminal
File Edit View Search Terminal Help
rpcclient $> enumprivs
found 35 privileges
SeCreateTokenPrivilege 0:2 (0x0:0x2)
SeAssignPrimaryTokenPrivilege 0:3 (0x0:0x3)
SeLockMemoryPrivilege 0:4 (0x0:0x4)
SeIncreaseQuotaPrivilege 0:5 (0x0:0x5)
SeMachineAccountPrivilege 0:6 (0x0:0x6)
SeTcbPrivilege 0:7 (0x0:0x7)
SeSecurityPrivilege 0:8 (0x0:0x8)
SeTakeOwnershipPrivilege 0:9 (0x0:0x9)
SeLoadDriverPrivilege 0:10 (0x0:0xa)
SeSystemProfilePrivilege 0:11 (0x0:0xb)
SeSystemtimePrivilege 0:12 (0x0:0xc)
SeProfileSingleProcessPrivilege 0:13 (0x0:0xd)
SeIncreaseBasePriorityPrivilege 0:14 (0x0:0xe)
SeCreatePagefilePrivilege 0:15 (0x0:0xf)
SeCreatePermanentPrivilege 0:16 (0x0:0x10)
SeBackupPrivilege 0:17 (0x0:0x11)
SeRestorePrivilege 0:18 (0x0:0x12)
SeShutdownPrivilege 0:19 (0x0:0x13)
SeDebugPrivilege 0:20 (0x0:0x14)
SeAuditPrivilege 0:21 (0x0:0x15)
SeSystemEnvironmentPrivilege 0:22 (0x0:0x16)
SeChangeNotifyPrivilege 0:23 (0x0:0x17)
SeRemoteShutdownPrivilege 0:24 (0x0:0x18)
Menu
rpcclient -U Hand 192....



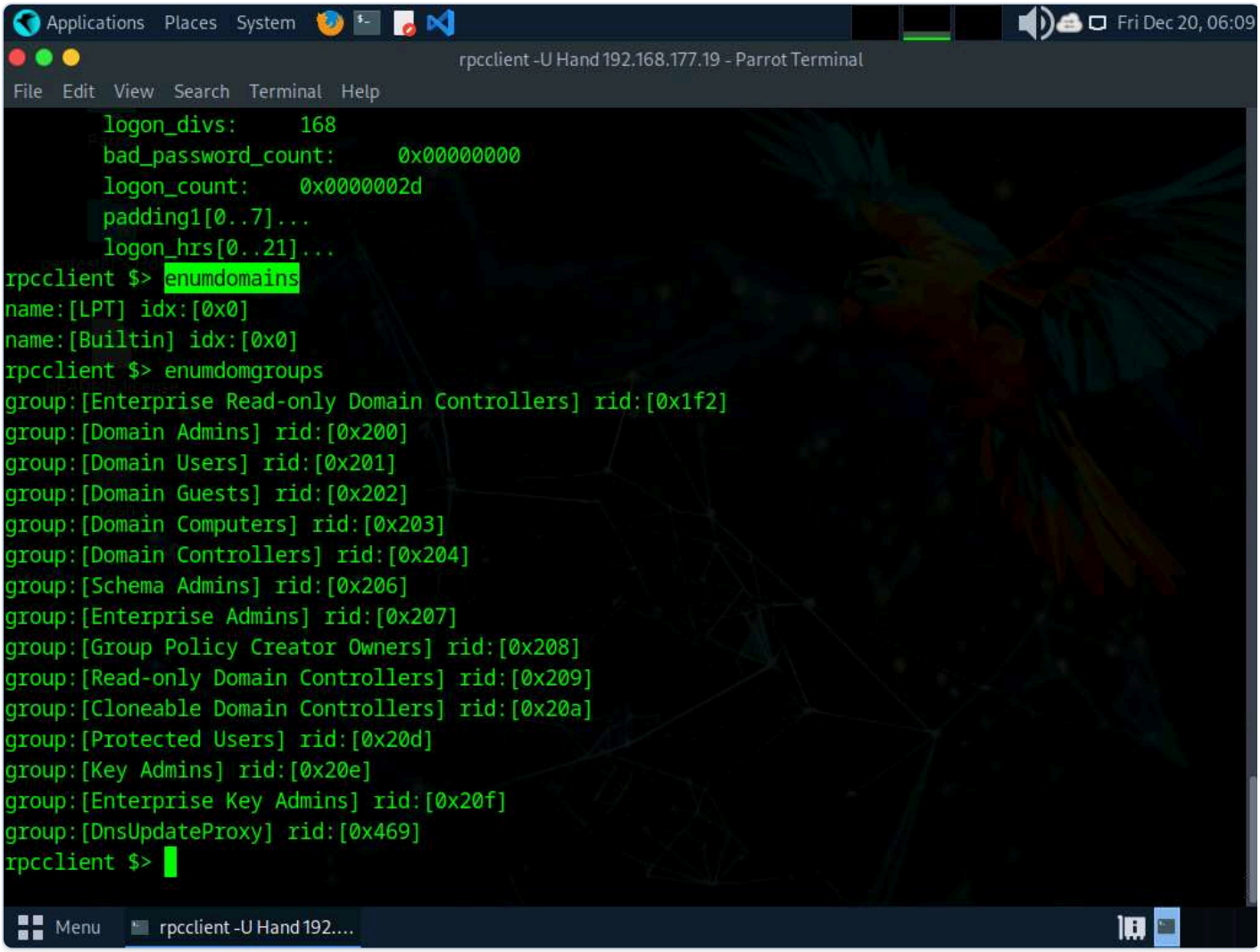
1. Query detailed account properties for user `Hand` using RID `0xc3a`:
   queryuser 0xc3a



1. Query detailed account properties for `Administrator` using RID `0x1f4`:
   queryuser 0x1f4

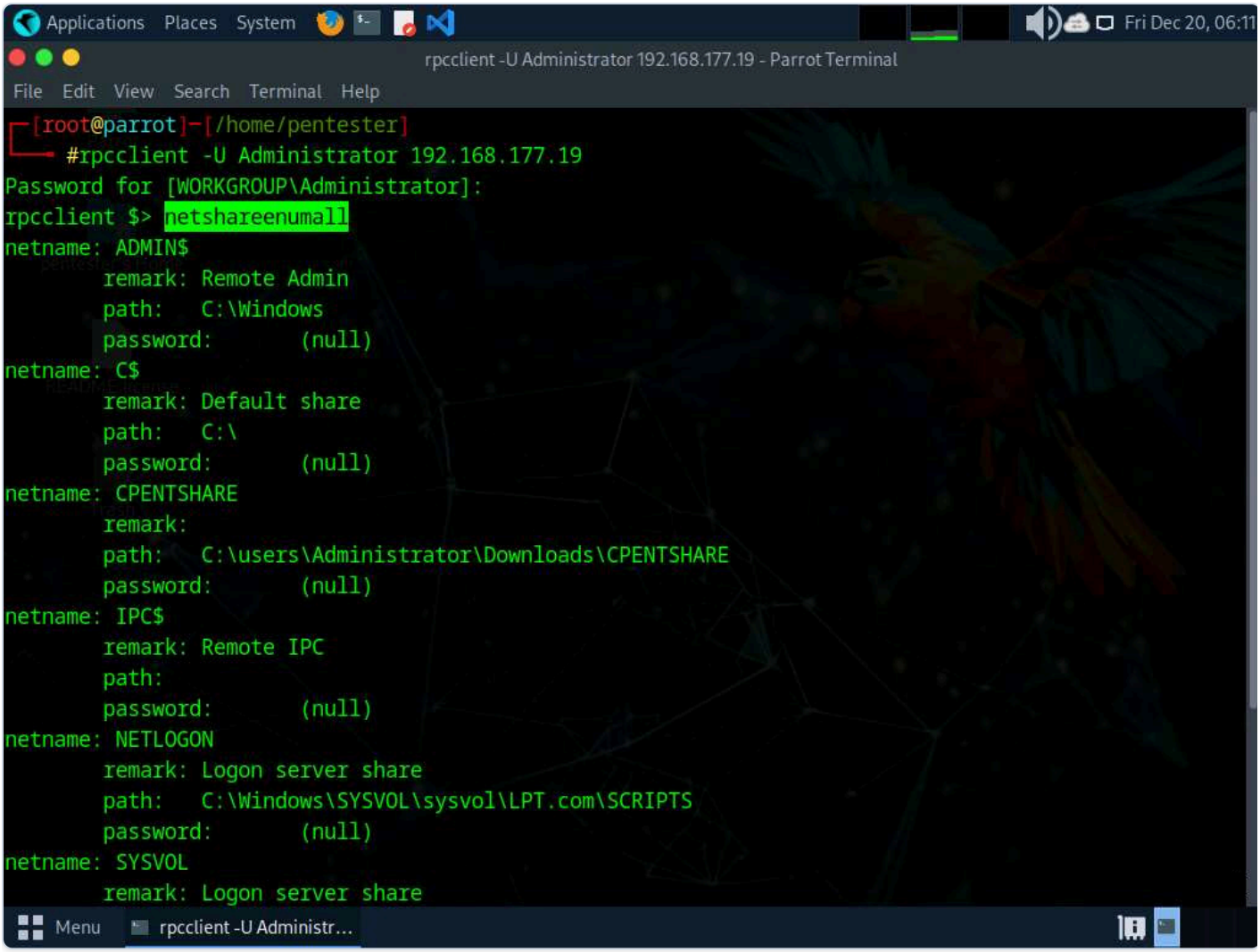


1. Enumerate domain groups and domain list:
   enumdomains
   enumdomgroups

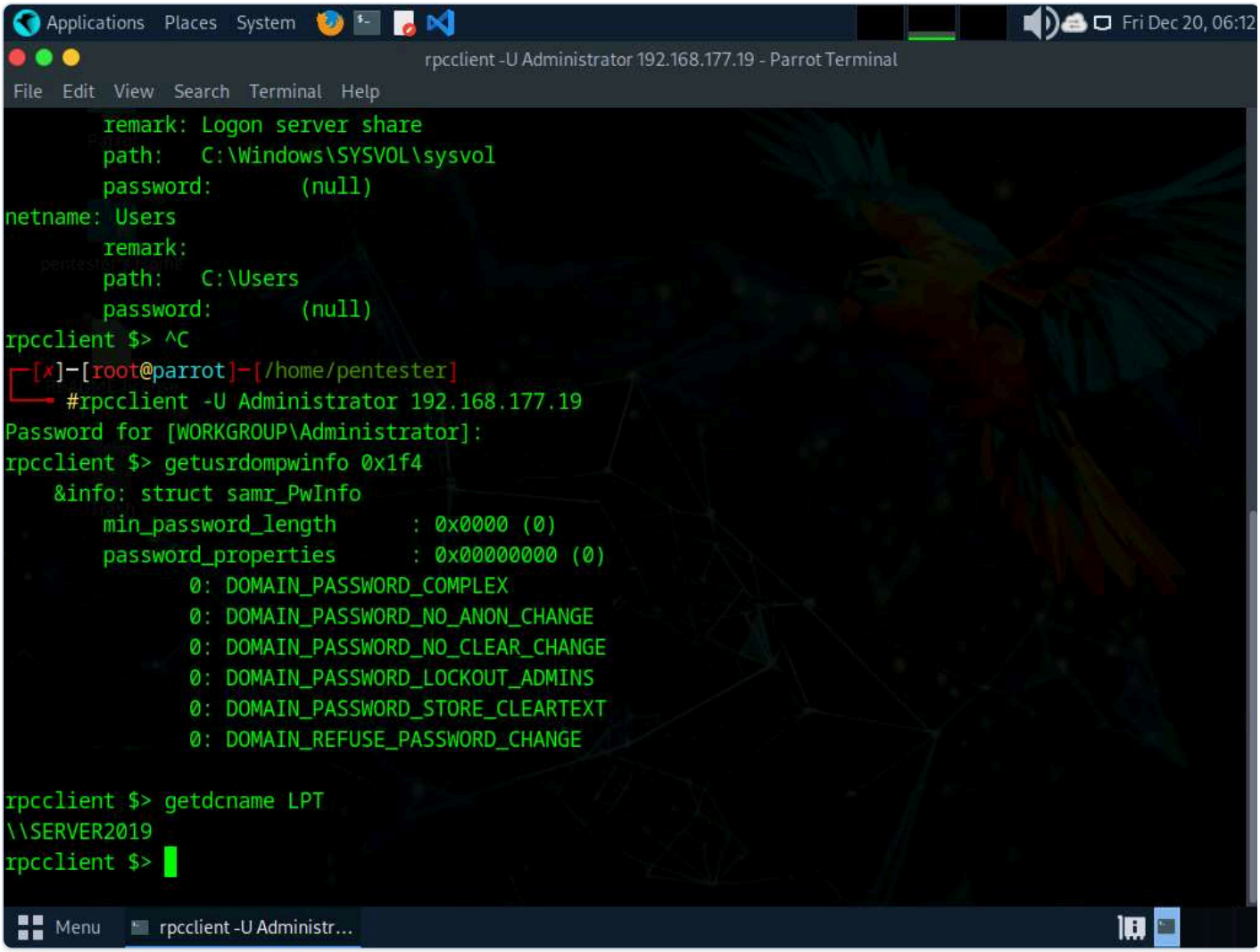


1. Open a new terminal and connect as Administrator:
   rpcclient -U Administrator 192.168.177.19
   Password: ***Pa$$w0rd***

2. Enumerate all network shares hosted on the target machine:
   netshareenumall



1. Extract domain password policy information and retrieve primary Domain Controller name:
   getusrdompwinfo 0x1f4
   getdcname LPT



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 9.3.1.1** On the Parrot Security machine, utilize the rpcclient tool to conduct RPC client enumeration. This process involves retrieving domain user details, privileges, shared resources, password policies, and domain information using various commands to extract valuable data from the target machine at 192.168.177.19. Enter the RID of the user jdoe (include square brackets). [!SUCCESS] ✅ **Answer:** **[0xe3c]**